

11. THE PIGEONHOLE PRINCIPLE

The pigeonhole principle states that if there are more pigeons than pigeonholes then there must be at least one pigeonhole having at least two pigeons in it.

In the set theoretic notation, the pigeonhole principle can be stated as follows:

If X and Y are any two finite sets such that $|X| < |Y|$ then a function $f: X \rightarrow Y$ can not be one-one i.e., there exists at least two elements x_1, x_2 in X such that $f(x_1) = f(x_2)$.

This principle is also applicable to other objects besides pigeons and pigeonholes.

Theorem 1. *If $(n + 1)$ or more objects are placed into n boxes (container) then there is at least one box containing two or more objects.*

Proof : Suppose no box of n boxes contains more than one object. Then the total number of objects will be n atmost. This is contradiction as there are at least $n + 1$ objects. Hence at least one box contains two or more objects.

Example 1. In a group of 27 English words there must be at least two words beginning with the same letter as there are 26 letters in English alphabet.

Example 2. In a group of 367 people there must be at least two people with the same birthday as there are 365 or 366 days in a year.

Example 3. In a class of 51 students there must be at least two students scoring same marks in a paper having maximum marks 50.

Example 4. Show that among any group of five integers (not necessarily consecutive) there are two integers with the same remainder when divided by 4.

Solution : There are four possible remainders when an integer is divided by 4. Therefore the pigeonhole principle implies that given five integers at least two have the same remainder.

Example 5. Show that at least in any set of seven classes there must be two that meet on the same day assuming that no class holds on Sunday.

Solution : There are seven classes and six days. Therefore by the pigeonhole principle there must be at least two classes holding on the same day.

Example 6. Show that among 13 persons there are at least two persons who were born in the same month.

Solution : If we consider 13 persons as pigeons and 12 months (Jan. . Dec.) as pigeonholes then by the pigeonhole principle these are at least persons having the birth in the same month.

Example 7. If 20 candidates appear in a competitive examination then show that there exists at least two among them whose Roll numbers differ by a multiple of 19.

Solution : Let X be the set of Roll numbers of 20 candidates and Y be the set of remainder when any positive integer is divided by 19
 $Y = \{0, 1, 2, \dots, 18\}$

Now we define a function $f: X \rightarrow Y$ as $f(x)$ = the remainder x is divided by 19

Since $|X| = 20$ and $|Y| = 19$ i.e., $|X| > |Y|$, therefore by the pigeonhole principle function f can not be one-one. This shows that there exists two distinct Roll numbers x_1 and x_2 such that

$$f(x_1) = f(x_2).$$

x_1, x_2 can be written as

$$x_1 = 19n_1 + f(x_1)$$

and

$$x_2 = 19n_2 + f(x_2)$$

where n_1 and n_2 are positive integers.

This gives

$$x_1 - x_2 = 19(n_1 - n_2)$$

$$[\because f(x_1) = f(x_2)]$$

Now $n_1 - n_2$ is an integer. Therefore $x_1 - x_2$ is a multiple of 19.

Hence there are at least two candidates whose Roll numbers differ by a multiple of 19.

Example 8. Determine the minimum number of elements to be selected from the set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that sum of two of them is 10.

Solution : The possible cases of selecting two elements from S whose sum is 10 will be as given below :

$$\{1, 9\}, \{2, 8\}, \{3, 7\}, \{4, 6\} \text{ and } \{5\}.$$

From these cases, we see that each element of S belongs to one and only one of these cases. If we select any two numbers from S then by the pigeonhole principle two of them will belong to the same set giving the sum equal to 10. Thus the required number to be selected from S is 6.

12. THE GENERALIZED PIGEONHOLE PRINCIPLE

Theorem 2. If N objects are placed into n boxes then there is at least one box containing at least $\left\lceil \frac{N}{n} \right\rceil$ objects, where $\lceil x \rceil$ denotes the largest integer less than or equal to x .

Let none of the boxes contains more than $\left\lfloor \frac{N}{n} \right\rfloor - 1$ objects. Then the total number of objects is at most.

$$n \left(\left\lfloor \frac{N}{n} \right\rfloor - 1 \right) < n \left(\left\{ \left\lfloor \frac{N}{n} \right\rfloor + 1 \right\} - 1 \right) = N \quad \left[\because \left\lfloor \frac{N}{n} \right\rfloor < \frac{N}{n} + 1 \right]$$

This is a contradiction because there are N total objects.

Example 9. Among 1000 people there are at least $\left\lfloor \frac{1000}{12} \right\rfloor = 84$ who were born in the same month.

Example 10. Show that there are at least four people in a city (population 25 million) with the same three initials who were born on the same day of the year (not necessarily in the same year).

Solution : There are 6,432,816 possibilities for the same three initials and a birthday. Therefore by the generalized pigeonhole principle there are at least $\lceil 25000,000/6432816 \rceil = 4$ people who share the same initials and birthday.

Example 11. What is the minimum number of students required in a class to be sure that at least five will receive the same grade if there are four possible grades A, B, C and D ?

Solution : The minimum number of students needed to ensure that at least 5 students receive the same grade is the smallest integer N such that $\left\lfloor \frac{N}{4} \right\rfloor = 5$. Such smallest integer is $N = 4 \cdot 5 + 1 = 17$. Thus 17 is the minimum number of students needed to ensure that at least 5 students will receive the same grade.

Example 12. In a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.

Solution : Let A be one of the six people of the group. There are either three or more among the other five people who are friends of A or enemies of A. This follows from the generalized pigeonhole principle as when five objects are divided into two sets, one has at least $\left\lceil \frac{5}{2} \right\rceil = 3$ objects. Let B, C and D are friends of A. If any two of these individuals are friends then these two and A form a group of three mutual friends. Otherwise B, C and D form a set of three mutual enemies. In the second case suppose B, C and D are enemies of A. If any two of these individual are enemies then these two and A form a group of three mutual enemies otherwise B, C and D form a group of three mutual friends.

Example 6. What is the least number of area codes needed to ensure that the 25 million phones in a state have distinct 10-digit telephone numbers (Assuming that telephone numbers are of the form $NXX-NXX-XXXX$ where first three digits form the area code, N represents a digit from 2 to 9 and X represents any digit).

Solution : By the product rule there are $8 \times 10 \times 10 = 800$ area codes with format NXX . The product rule also shows that there are $10 \times 10 \times 10 \times 10 = 10,000$ station codes with format $XXXX$. Therefore there are $800 \times 10,000 = 8,000,000$ different phone numbers of the form $NXX-XXXX$. Hence by the generalized pigeonhole principle among 25 million telephones at least $\lceil 25,000,000/8,000,000 \rceil = 4$ of them must hence identical phone numbers. Hence at least four area codes are required to ensure that 10 digit numbers are different.

Example 7. Show that among any $(n + 1)$ positive integers not exceeding $2n$ there must be an integer that divides one of the other integers.

Solution : We write each of the $(n + 1)$ integers $a_1, a_2, a_3, \dots, a_{n+1}$ as a power of 2 times an odd integer i.e. $a_i = 2b_i$ for $i = 1, 2, \dots, (n + 1)$ where i is a non-negative integer and b_i is odd integer. Here $b_1, b_2, \dots, b_{n+1}$ are all odd positive integers less than $2n$. Since there are only n odd positive integers less than $2n$ it follows from the pigeonhole principle that two of the integers $b_1, b_2, \dots, b_{n+1}$ must be equal. Therefore there are integers i and k such that $b_i = b_k$. Let b be the common value of b_i and b_k . Then $a_i = 2^j b$ and $a_k = 2^l b$. Now if $j_1 < j_2$ then obviously a_{j_1} divides a_{j_2} and if $j_1 > j_2$ then a_{j_2} divides a_{j_1} .

14. The Principle of Inclusion-Exclusion : When two jobs can be done at the same time we can not use the sum rule to count the number of ways to do one of the two jobs. Adding the number of ways to do each job leads to an overcount since the ways to do both jobs are counted twice. Therefore to count the number of ways correctly to do one of the job we add the number of ways to do each of the two jobs and then subtract the number of ways to do both jobs. This technique is known as the principle of inclusion-exclusion.

Theorem 1. (The Principle of Inclusion-Exclusion). If $A_1, A_2, \dots, A_n$ are finite sets then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{i=1}^n |A_i| - \sum_{i,j=1}^n |A_i \cap A_j| + \sum_{i,j,k=1}^n |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} \sum |A_1 \cap A_2 \cap \dots \cap A_n|$$

where $|A_i|$ is the number of elements of A_i .

Example 1. Give a formula for the number of elements in the union of three sets.

Solution : The inclusion-exclusion principle gives that

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_2 \cap A_3| - |A_1 \cap A_3| + |A_1 \cap A_2 \cap A_3|.$$

Example 2. How many positive integers not exceeding 500 are divisible by 7 or 11 ? (Meerut 2003)

Solution : Let A be the set of positive integers not exceeding 500 that are divisible by 7 and let B be the set of positive integers not exceeding 500 that are divisible by 11. Then $A \cup B$ is the set of integers not exceeding 500 that are divisible by 7 or 11 and $A \cap B$ is the set of integers not exceeding 500 that are divisible by both 7 and 11. Now

$$\begin{aligned}|A \cup B| &= |A| + |B| - |A \cap B| \\&= \left\lfloor \frac{500}{7} \right\rfloor + \left\lfloor \frac{500}{11} \right\rfloor - \left\lfloor \frac{500}{7 \cdot 11} \right\rfloor \\&= 71 + 45 - 6 \\&= 110.\end{aligned}$$

Thus there are 110 positive integers, not exceeding 500 that are divisible by 7 or 11.

Example 3. A discrete mathematics class contains 30 students majoring in computer science, 18 students majoring in mathematics and 8 students majoring in computer science and mathematics both. How many students are in this class if every student is majoring in mathematics, computer science or both.

Solution : Let A be set of students in the class majoring in computer science and E be the set of students majoring in mathematics. Then $A \cap B$ is the set of students in the class majoring in computer science and mathematics both. Since every student in the class is majoring in either computer science or mathematics or both it follows that the number of students in the class is $|A \cup B|$. Now

$$\begin{aligned}|A \cup B| &= |A| + |B| - |A \cap B| \\ &= 30 + 18 - 8 \\ &= 40\end{aligned}$$

Example 4. A total of 1200 students have taken a course in Hindi, 800 have taken a course in English and 140 have taken Urdu. Further 100 have taken course in both Hindi and English, 30 have taken in both Hindi and Urdu and 20 have taken both English and Urdu. If 2000 students have taken at least one language, how many students have taken a course in all the three languages ?

Solution : Let H be the set of students who have taken course in Hindi, E be the set of students who have taken a course in English and U be the set of students who have taken a course in Urdu. Then

$$\begin{aligned}|H| &= 1200, |E| = 800, |U| = 140 \\ |H \cap E| &= 100, |H \cap U| = 30, |E \cap U| = 20 \\ \text{and } |H \cup E \cup U| &= 2000.\end{aligned}$$

Now by the principle of inclusion-exclusion we have

$$\begin{aligned}|H \cup E \cup U| &= |H| + |E| + |U| - |H \cap E| - |H \cap U| \\ &\quad - |E \cap U| + |H \cap E \cap U|\end{aligned} \quad \dots(1)$$

Putting the above values in (1) we get

$$\begin{aligned}2000 &= 1200 + 800 + 140 - 100 - 30 - 20 + |H \cap E \cap U| \\ \therefore |H \cap E \cap U| &= 10.\end{aligned}$$

Therefore there are ten students who have taken course in Hindi, English and Urdu.

Example 5. At an art academy there are 45 students taking ceramics, 55 students taking painting and 30 students taking sculpture. There are 15 students in both ceramic and painting, 10 in both painting and sculpture, 5 in both ceramic and sculpture and 2 taking all the three courses. How many students are taking at least one course at the art academy ?

Solution : Let C , P and S denote the sets of students taking ceramics, painting and sculpture respectively. We have to calculate $|C \cup P \cup S|$. Applying principle of inclusion-exclusion, we have

$$\begin{aligned}|C \cup P \cup S| &= |C| + |P| + |S| - |C \cap P| \\ &\quad - |C \cap S| - |P \cap S| + |C \cap P \cap S| \\ &= 45 + 55 + 30 - 15 - 10 - 5 + 2 \\ &= 102.\end{aligned}$$

Thus 102 students are taking at least one course at the art academy.

Example 6. How many solution does

$$x_1 + x_2 + x_3 = 13$$

have where x_1, x_2, x_3 are non-negative integers with $x_1 \leq 4, x_2 \leq 5$ and $x_3 \leq 6$.

Solution : Let $x_1 > 4$ be the property P_1, P_2 and P_3 be the properties of $x_2 > 5$ and $x_3 > 6$ respectively. Then the number of solutions satisfying the inequalities $x_1 \leq 4, x_2 \leq 5$ and $x_3 \leq 6$ is given by

$$N(P_1' P_2' P_3') = N - N(P_1) - N(P_2) - N(P_3) + N(P_1 P_2) + N(P_2 P_3) + N(P_1 P_3) - N(P_1 P_2 P_3) \quad \dots(1)$$

Now

$$\begin{aligned} N &= \text{total number of solutions} \\ &= C(3 + 13 - 1, 13) = C(15, 13) \\ &= \frac{15 \times 14}{2 \cdot 1} = 105 \end{aligned}$$

$$\begin{aligned} N(P_1) &= (\text{number of solutions with } x_1 \geq 5) \\ &= C(3 + 8 - 1, 8) = C(10, 8) = 45 \end{aligned}$$

$$\begin{aligned} N(P_2) &= (\text{number of solutions with } x_2 \geq 6) \\ &= C(3 + 7 - 1, 7) = C(9, 7) = 36 \end{aligned}$$

$$\begin{aligned} N(P_3) &= (\text{number of solutions with } x_3 \geq 7) \\ &= C(3 + 6 - 1, 6) = C(8, 6) = 28 \end{aligned}$$

$$\begin{aligned} N(P_1 P_2) &= (\text{number of solutions with } x_1 \geq 5 \text{ and } x_2 \geq 6) \\ &= C(3 + 2 - 1, 2) = C(4, 2) = 6 \end{aligned}$$

$$\begin{aligned} N(P_2 P_3) &= (\text{number of solutions with } x_2 \geq 6 \text{ and } x_3 \geq 7) \\ &= C(3 + 0 - 1, 0) = C(2, 0) = 1 \end{aligned}$$

$$\begin{aligned} N(P_1 P_3) &= (\text{number of solutions with } x_1 \geq 5 \text{ and } x_3 \geq 7) \\ &= C(3 + 1 - 1, 1) = C(3, 1) = 3 \end{aligned}$$

$$N(P_1 P_2 P_3) = (\text{number of solutions with } x_1 \geq 5 \text{ and } x_2 \geq 6 \text{ and } x_3 \geq 7) = 0$$

Putting these values in (1) we get that the number of solutions with $x_1 \leq 4, x_2 \leq 5$ and $x_3 \leq 6$ is given by